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EEG Analysis based on Spectral Edge Frequency during Mental Arithmetic Task

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***Abstract*— Brain dynamics variables and features are critical in electroencephalogram (EEG)-based human cognition research. They allow researchers to investigate how functionality changes based on the environment and how a task is carried out. This work gives a unique perspective of EEG analysis based on spectral edge frequency 50% (SEF50) methodologies to examine the differentiation of brain electrical activity during mental arithmetic task based on two circumstances (a) rest and task; and (b) mental arithmetic task success rate. According to our findings, the spectral edge frequency of the signals is increasing in beta and gamma-wave activity during mental arithmetic activities compared to rest state. In terms of performance, trends in the SEF50 curves indicate that average performers and bad performers likely expend more cognitive effort than good performers, particularly in higher-frequency ranges. Notably, the data revealed that frontal and pre-frontal brain regions indicate the brain activity compare to other brain regions. In conclusion, the SEF50 techniques used in this research are insightful and can be regarded as a feature of EEG data.**

Keywords— SEF50, mental arithmetic, EEG, band frequency, brain region

I. Introduction

The topic about human cognitive performance has attracted a lot attention of many researchers due to its active field of theoretical and practical investigation [1]-[7]. One of cognitive workload that commonly used is a mental arithmetic. The complexity and variability of mental arithmetic tasks offer a rich landscape for cognitive load modulation, making them a choice for a study like ours. It encompasses key cognitive functions including working memory, concentration, and numerical processing. These abilities can be seen from brain electrical signals activity with electroencephalogram (EEG) [1].

EEG is a powerful tool based on non-invasive approach to measure brain activity. It offers high temporal resolution, enabling scientists to examine the rapid changes of activation from brain areas in cognitive states that other neuroimaging techniques might miss. To further enrich the interpretive power of EEG data, numerous analytical measures have been developed, such as power spectral density, fast Fourier transform, coherence, and spectral edge frequency (SEF) [7]-[17].

SEF can be employed to examine the cerebral cortex, and it signals the level of workload and depth of anesthesia [17][18]. SEF quantitatively interprets the area ratio against the total frequency range in Hz, demonstrating the extent to which the ratio is biased towards high frequencies and the distribution of the power spectrum of brain waves [19][20]. SEF comes in several forms, such as SEF 25%, SEF 50% (median frequency), SEF 90%, and SEF 95% [17]. However, SEF50 is employed to evaluate the brain's activity while executing a task in an alert state [18][21]. Furthermore, SEF50 can be an indicator for cognitive load in the brain; the cognitive ability improves as the SEF50 value increases [22].

Numerous studies have been published on EEG analysis for cognitive stress utilizing Physionet mental arithmetic datasets [23]. The methodologies employed in these analyses comprise: (a) detrend fluctuation, coherence, and spectral power analysis [7]; (b) fractal dimension and power spectral density [24]; and (c) functional brain connectivity [25]. However, in mental arithmetic research, no studies have been undertaken an EEG analysis of mental arithmetic using the SEF method. Therefore, this paper provides a distinctive view of mental arithmetic through SEF50 lens. The analysis involves observing the differences in the EEG signals of subjects during two conditions as in [7][24]: (a) rest and task; and (b) based on the arithmetic success rate of the subjects.

Despite the individual popularity of SEF 50% and mental arithmetic tasks in neuroscience, their combined use is yet to be fully explored. This paper aims to bridge this gap by leveraging SEF 50% as a neurophysiological marker of cognitive dynamics during mental arithmetic tasks. Our findings show that SEF50 effectively embodies the changes in cognitive load during mental calculation. Through the perspective of SEF, we aim to enhance our understanding of the neural dynamics involved in mental arithmetic, thereby contributing to the wider objective of understanding human cognition and its neural mechanism.

The rest of this paper is structured into four parts: an introduction, a description of the materials and methods, a section on results and discussion, and a conclusion. The

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introduction provides an overview of the research background. The materials and methods section outlines the dataset used in this study and the techniques employed for data analysis. The results and discussion section showcase the findings derived from data processing and its subsequent analysis. The final section, the conclusion, summarizes the findings from the data processing and analysis and hints at potential future works.

II. Materials and Methods

A. Dataset

A detailed explanation of the data used in this research is available in Igor Zyma et al. [23]. The dataset was sourced from the PhysioBank dataset. EEG were captured using the Neurocom monopolar EEG 23-channel system. Location of electrodes placement according the International 10/20 scheme are temporal lobe (T3, T4, T5, T6), frontal lobe (F3, F4, Fz, F7, F8), and parietal lobe (P3, P4, Pz). Electrodes also placed in pre-frontal (Fp1, Fp2), central (C3, C4, Cz), and occipital (O1, O2) as illustrated in Fig. 1. All electrodes were connected to interconnected ear reference electrodes with an inter-electrode impedance of less than 5 kΩ. 500 Hz was the sample rate for each channel. A high-pass filter with a cut-off frequency of 0.5 Hz, a low-pass filter with a cut-off frequency of 45 Hz, and a power line notch filter (50 Hz) were used; the time constant of the amplification tract was 0.3 seconds.

There were 66 healthy, right-handed participants in the study, with an average age of 18.6 years and an SD of 0.08 years. Participants performed a serial mental arithmetic task involving sequential subtraction of a 4-digit and 2-digit number. Each participant's general success (good) or failure (bad) was determined by the number of operations completed during a 4-minute period and their success rate score, which was determined by the accuracy of their calculations. Each participant's EEG recording consisted of 180 seconds of rest (rest) and 60 seconds of arithmetic workload (task).

Due to substandard EEG quality, 30 of the initial 66 participants were excluded from the database, leaving a final sample of 36 subjects. On the basis of calculation success, these 36 volunteers were categorized as either good (G) or poor (B) counters: the G group comprised 24 subjects with an average of 24 (SD=7.4) arithmetic operations, while the B group comprised 12 subjects with an average of 7 (SD=3.6) arithmetic operations.

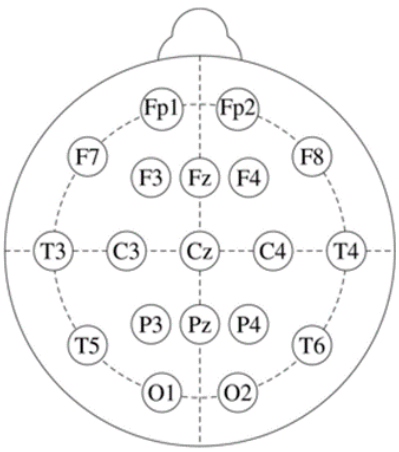

Fig. 1. The electrode placement.

B. Methods

a) SEF50

SEF at 50% (SEF50) is the frequency below which fifty percent of the signal's power is present [26]. This is equivalent to the signal's median frequency. It is derived from the power spectral density (PSD) according to (1).

$$PSD\ (f) = \lim_{T\to\infty} \frac{1}{T} |\hat{x}_T(f)|^2 \tag{1}$$

where $|\hat{x}_T(f)|^2$ can be seen as the Fourier transform of the time convolution and T is a period of the time convolution.

After we obtained the PSD value, we calculated the total area of PSD as in (2).

$$total\ area = \int [f_{min}\ to\ f_{max}]\ PSD\ (f) df \tag{2}$$

Last, we can obtain the value of SEF50 by (3)

$$SEF50 = 0.5 \times\ total\ area \tag{3}$$

The example of SEF50 calculation that was completed for this study is shown in Fig. 2.

b) EEG Processing

Fig. 3 shows the block diagram of EEG processing for this study. In this study, EEG signal processing was performed for 36 subjects across all channels and all EEG frequency bands. The process began by applying a bandpass filter and a notch filter to the raw EEG signal. Then, for each channel of each subject, extraction was conducted to obtain the respective frequency bands of the EEG signal. The respective frequency band used in this study is divided into five range: delta (0.5 Hz – 4 Hz), theta (4 Hz – 8 Hz), alpha (8 Hz – 13 Hz), beta (13 Hz – 33 Hz), and gamma (33 Hz – 44 Hz). Subsequently, SEF50 calculations were performed for each of these frequency bands. Once the SEF50 values were obtained, the average SEF values for each frequency band were calculated based on brain regions. The brain regions are divided into six parts: pre-frontal (Fp), frontal (F), central (C), parietal (P), occipital (O), and temporal (T).

After obtaining the average value of SEF50 for each brain region from each frequency band, further analysis was conducted for two conditions: (a) rest and task; and (b) based on arithmetic success rate. For the second condition, we followed the grouping of mental arithmetic success levels as outlined in paper [24], which is divided into three groups: good, average, and poor performers, with the name of the group reflecting the subjects' performances. Each group consists of 12 participants according to the rank of the number of correct subtractions they achieved during the test.

III. Result and Discussion

First, we present the analysis of SEF50 during the rest and task state of subjects in section III.A. Next, in section III.B, we explore the analysis of SEF50 among performance group.

A. Analysis of Rest and Task State

The initial set of findings displays the average value of SEF50 from 36 participants, sorted by brain region and frequency band, as depicted in the individual graphs of Fig.4.

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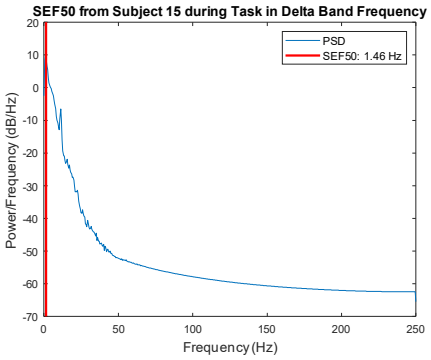

Fig. 2. The example of SEF50.

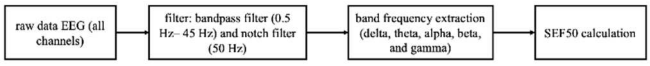

Fig. 3. The block diagram of EEG processing.

Each graph contains two curves representing brain activity during a mental arithmetic task and while resting. Given the consistent scaling on each graph's axes, it's clear that the beta and gamma frequency bands are more distinct across all brain regions. As SEF50 values rise, a shift in the dominance between resting and task curves is visible, indicating an increased presence of beta and gamma waves during tasks. This pattern aligns with an active, problem-solving mindset, consistent with the mental arithmetic task. This observation is supported by paper [24] that evaluate the rest and task state using power spectral density, which also noted with the increasing of brain activity in the beta and gamma bands during the tasks. Notably, the highest SEF50 values appear in the pre-frontal (Fp) and frontal (F) areas. This aligns with the understanding that these brain regions are crucial for cognitive activities, decision-making, personality traits, and handling intricate cognitive challenges.

To clarify from Fig. 4, the difference in SEF50 values between rest and task can be observed in Table 1. The difference value is calculated using Eq. (4).

$$difference\ value = SEF50_{task} - SEF50_{rest} \quad (4)$$

For the delta, theta, and alpha frequency bands, there tends to be a decrease across all brain regions. In contrast, the beta and gamma frequency bands show an increase in all brain regions. The significance of these changes in the theta, beta and gamma bands can also be discerned from the subsequent T-test values ($p<0.05$), except for the beta band in frontal and central region. The decrease of SEF50 in theta band and the increase of SEF50 in beta and gamma band indicate that the subjects have reached the stage of concentration state and reduced the relax state.

TABLE I. THE DIFFERENCE VALUE OF REST AND TASK

Band-Freq	Difference of Rest and Task in SEF50 (Hz)					
	Fp	*F*	*C*	*O*	*P*	*T*
Delta	0.041	0.022	-0.041	-0.081	-0.090	-0.003
Theta	-0.841	-0.648	-0.538	-0.787	-0.787	-0.587
Alpha	-0.237	-0.171	0.023	0.231	0.109	0.264
Beta	1.180	0.678	0.778	1.139	1.334	1.224
Gamma	2.109	1.902	1.682	2.184	1.994	1.953

B. Analysis of Performance Group

This segment focuses on discerning variations in the EEG signals of subjects according to their performance levels in the mental arithmetic task: good, average, and poor. Fig. 5 shows the average value of SEF50 for these three categories, segmented by brain region and frequency band. Primarily, our analysis emphasizes the subplots related to the beta and gamma bands. The reason of focusing only on these band is due to their relevance to areas of the brain associated with concentration and problem-solving, as identified earlier in section III.A.

In general, good performers have SEF50 values that are equivalent to or larger than average performers in the beta frequency throughout all brain areas. However, the peak of SEF50 values for good performers is never bigger than the highest of SEF50 values for poor performers. Furthermore, across all brain regions, the curve for good performers in the gamma frequency band tends to be lower than the others, particularly in this higher-frequency spectrum. SEF50 peak values occur in the frontal, pre-frontal, and temporal areas, with SEF50 peak values exceeding the 25 Hz mark. However, the curves of gamma frequency band for average and bad

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performers are raised across all brain regions compared to high performers, culminating in the frontal, pre-frontal, and temporal regions and exceeding the 27 Hz mark.

To put it simply, good performers only need to exert cognitive effort up to the beta frequency region, resulting in lower SEF50 levels in the gamma frequency band. Poor and average performers, on the other hand, may require greater concentration and cognitive effort during subtraction exercises than good performers, resulting in increased EEG activity at higher frequencies. This observation is supported by studies [24] and [25], in which the authors propose a similar pattern. Furthermore, for both conditions of analysis, namely (i) rest and task; (ii) arithmetic success rate, it is shown that the frontal and prefrontal regions reflect the brain activity. This is in accordance with [27] that both of these regions are responsible for the cognitive functions connected with mental arithmetic.

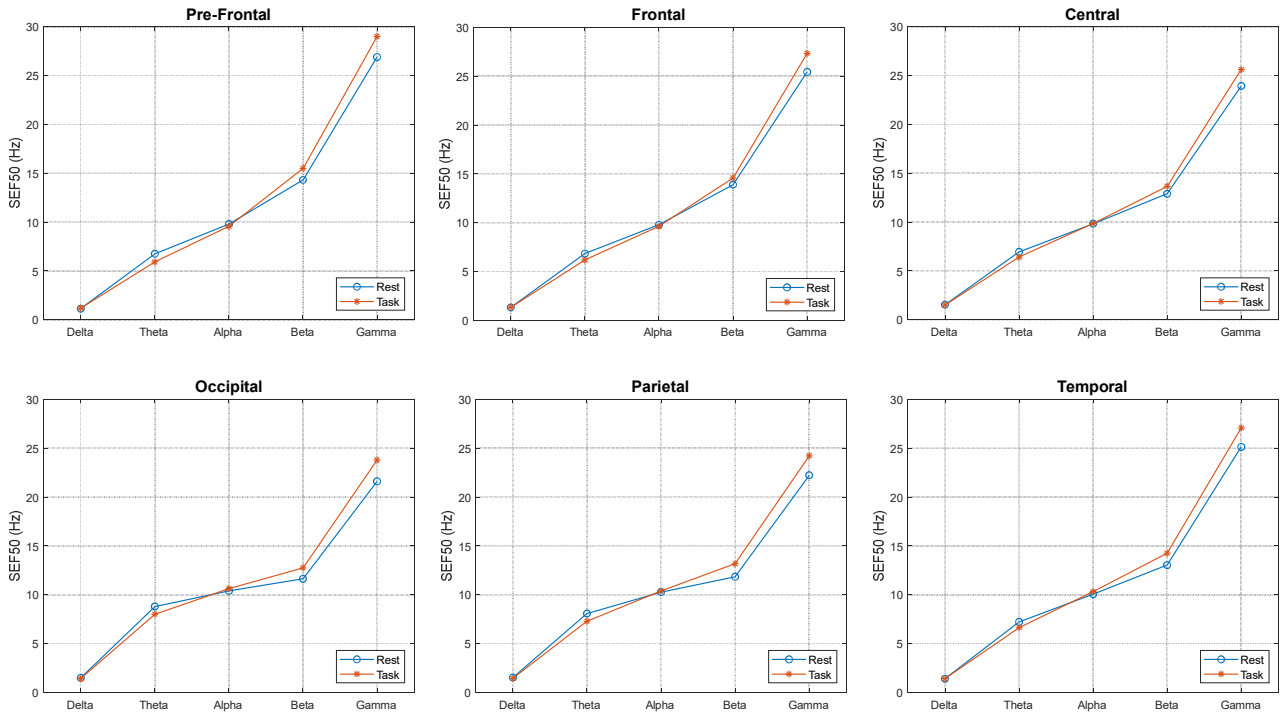

Fig. 4. Comparison of the SEF50 between task and rest datasets (considering the average for all 36 subjects).

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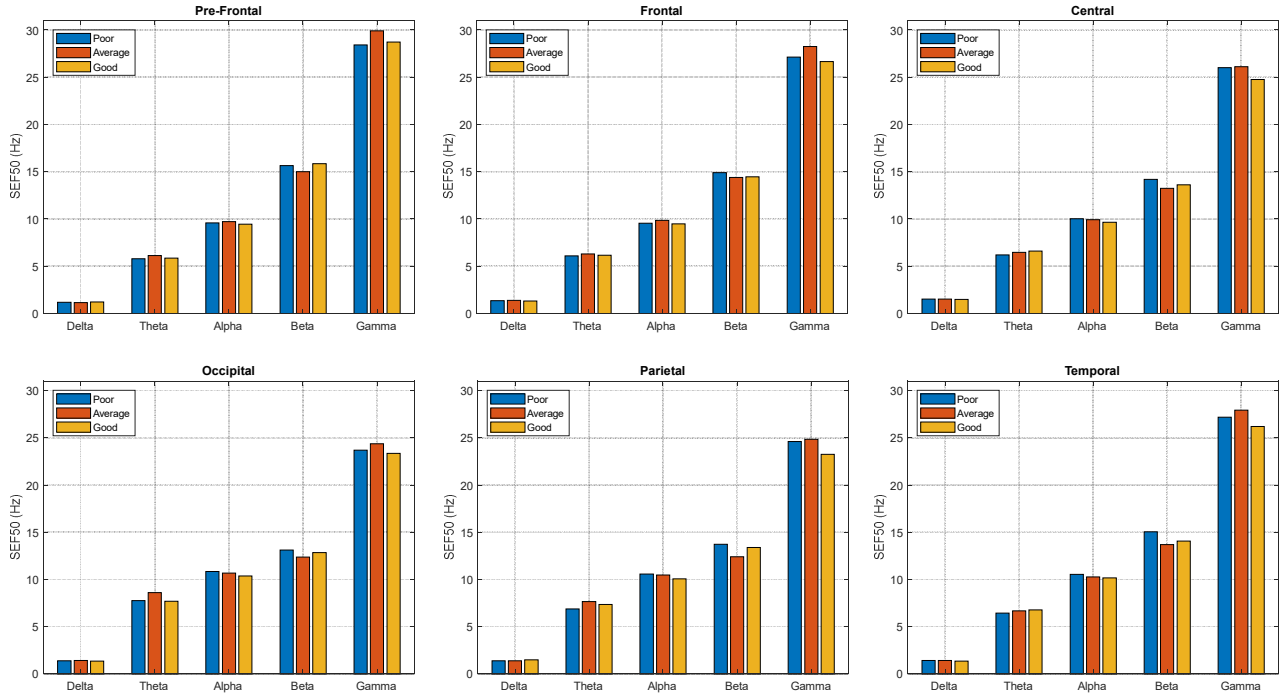

Fig. 5. Comparison of the SEF50 between good, average, and poor performers (considering the average for each group of 12 subjects).

IV. Conclusion

In this research, we utilized SEF50 techniques to analyze existing EEG data, aiming to identify patterns in subjects doing a mental arithmetic task both at rest and during activity. The purpose was also to assess their performance levels during these mental arithmetic tests. The EEG data was structured based on brain region and frequency band, presenting average SEF50 values derived from our EEG analysis.

Our examination of the spectral edge frequency within the SEF50 framework revealed increased beta and gamma-wave activity in subjects while they were actively engaged in mental arithmetic activities, compared to resting conditions. And also, the SEF50 patterns indicated that the average and bad performers appeared to exert more cognitive effort than the good performers, particularly in the higher frequency bands. Furthermore, the data also revealed that the frontal and prefrontal brain areas reflect the brain activity than the others of brain region.

To sum it up, the SEF50 technique proved to be valuable in our analysis and it can provide deeper insights into EEG feature related to mental arithmetic tasks. For a more comprehensive understanding, our future work will involve integrating EEG data with electrocardiogram (ECG) data available from the same mental arithmetic dataset.

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